Monday, June 1, 2026 8:46 PM

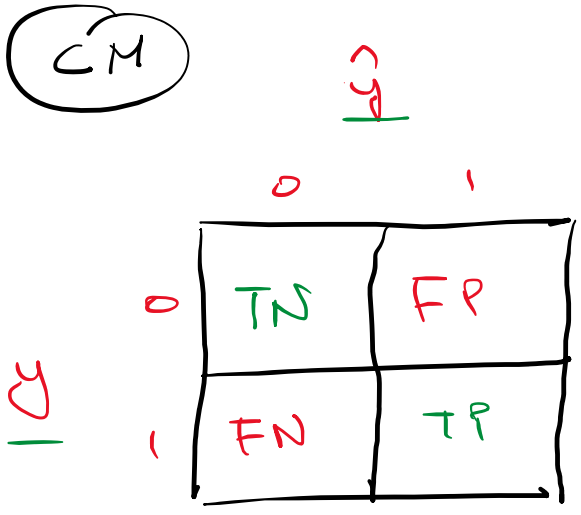
Classification METRICS & imBalanced DATA

Agenda ⇒

- (*) confusion matrix ⇒
  - → TPR | TNR | FPR | FNR
- (*)
  - → ROC | AUC CURVE
  - → imBalance DATA ⇒

(*) Recap ⇒

CM



(*) spam | NS

(*) Precision = TP / (TP + FP) ⇒ CORRECT

→ V. V. imp

(*) Recall = TP / (TP + FN)

(*) F1 score = $\frac{2 \cdot P \cdot R}{P + R}$

True +ve Rate

TPR | TNR → True -ve Rate

+ve → S
-ve → NS

sensitivity ⇒ +ve class

(+) TPR = $\frac{TP}{\text{actual all +ve}} = \frac{TP}{TP+FN}$ = Recall

-ve class

(*) TNR = $\frac{TN}{\text{actual all -ve}} = \frac{TN}{TN+FP}$

↳ specificity

NS

(*) Non spam emails ⇒ TNR (*)

↳ pred. as spam

(*) FPR ↓

④ $FPR = \frac{FP}{\text{all -ve}}$

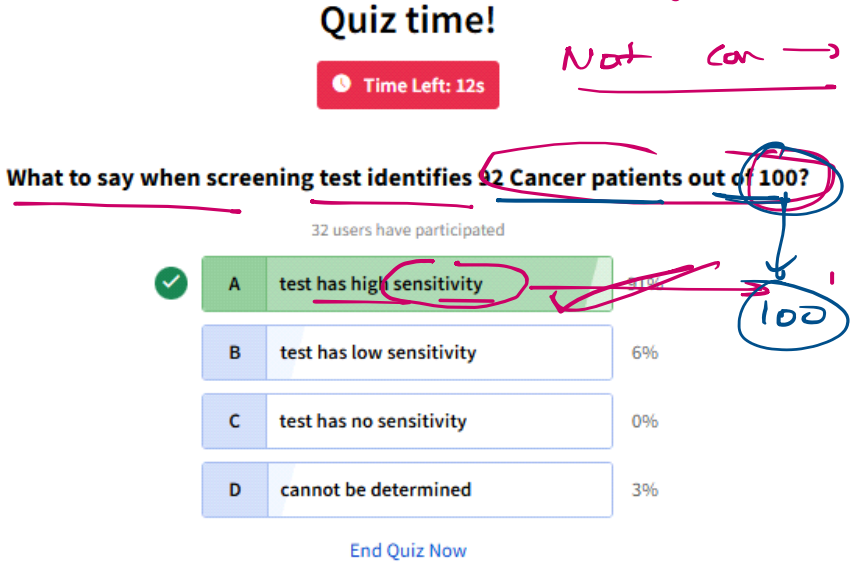
pred. as spam

$= \frac{FP}{TN + FP}$

⑤ $FNR = \frac{FN}{\text{all +ve}} = \frac{FN}{TP + FN}$

Cancer → +ve

Not can → -ve



92/100 → 92%

92 → Cancer

& → No Cancer

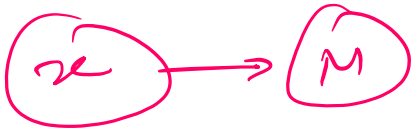
→ Physics concept

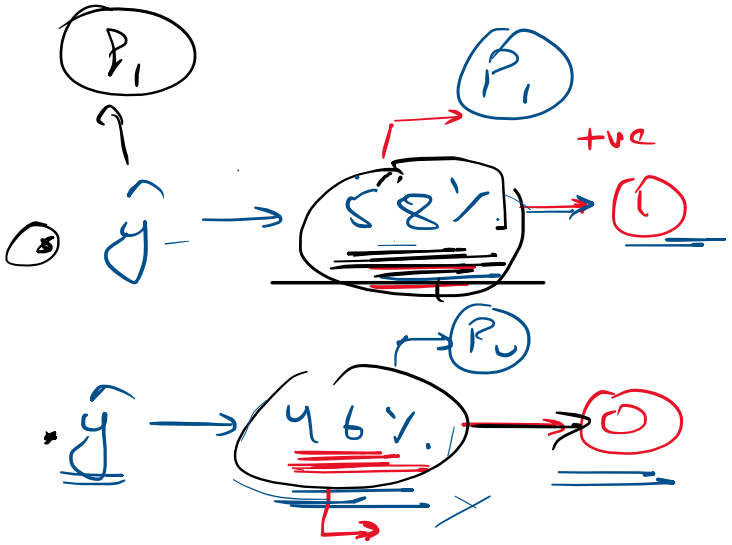
⑥ ROC (Receiver operating characteristic)

↳ ROC curve

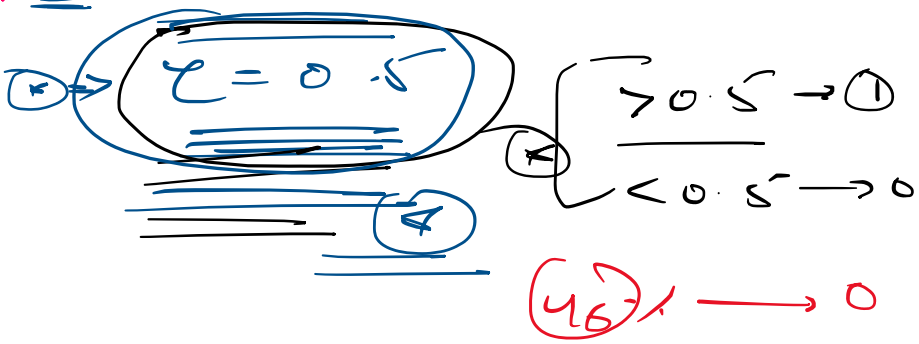
eval metric

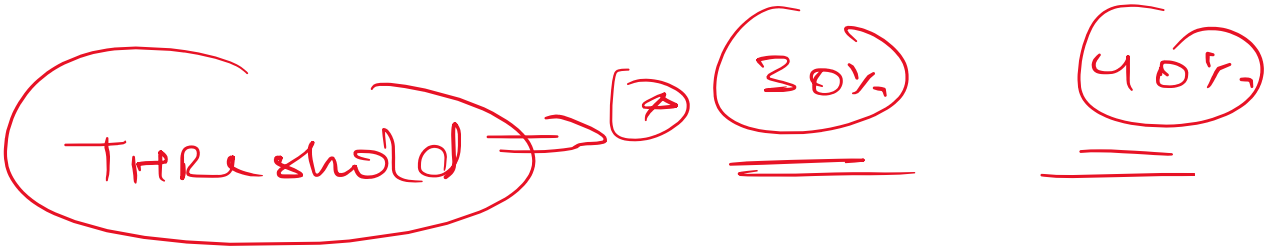
Scenario





$P(y=1 \mid x)$



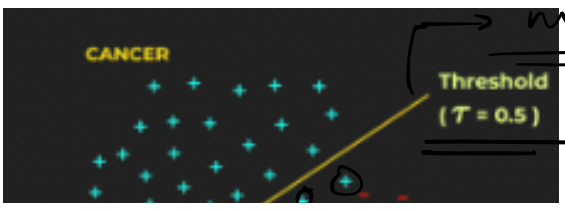


Changing Threshold ⇒

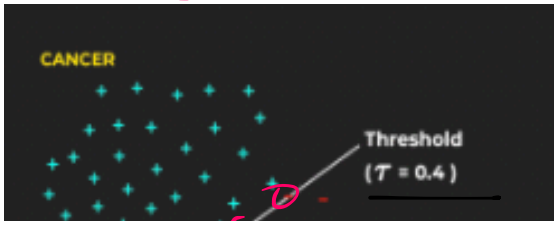
errors = 5

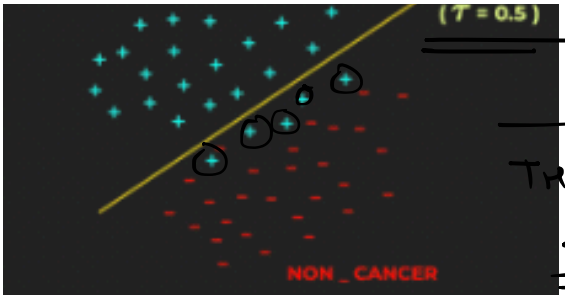
Cancer or No cancer

errors = 5

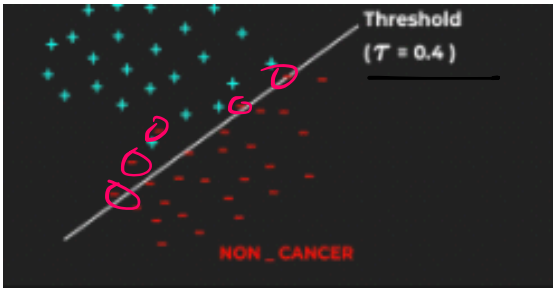


model





Threshold
smitpd



FN errors (I)

False +ve (2)

⇒ more dangerous ✗

⇒ Better case ✓

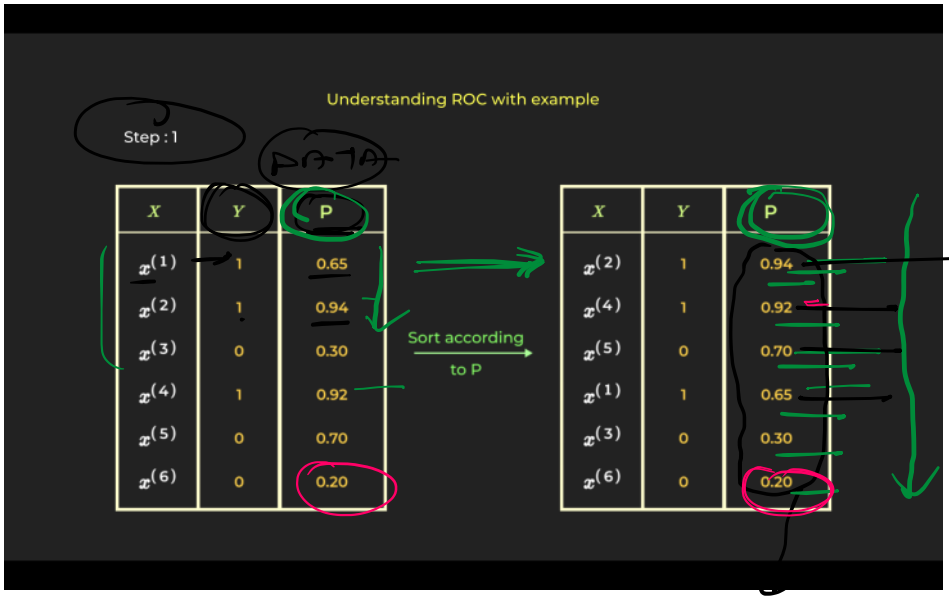
⇒ How do i find the Best Threshold

ROC → MC (Physic)

evaluate how good our classifier is

diff. THRESHOLDS ⇒ ROC Score

ROC process =>

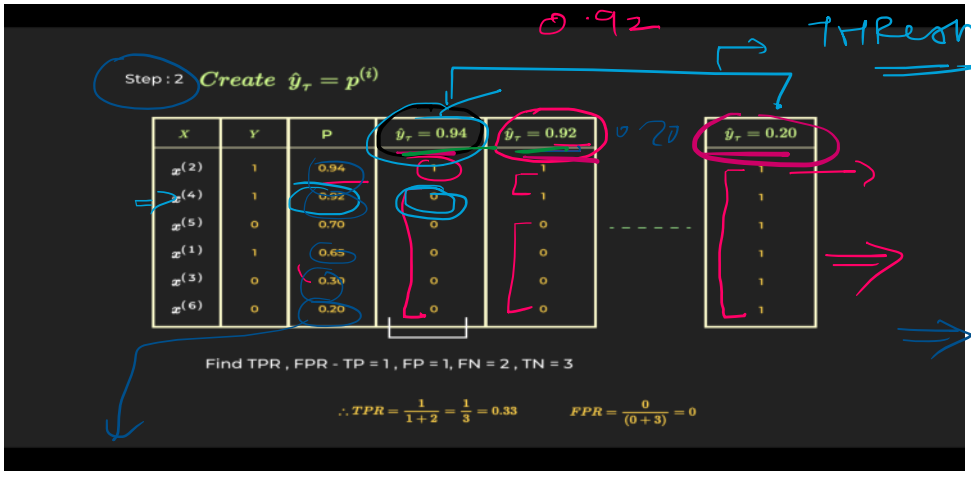


Reordering the data

Threshold → Predn

step 2 → (=1) 0.94 → prob ≥ 0.94 → (1)

otherwise → (0)

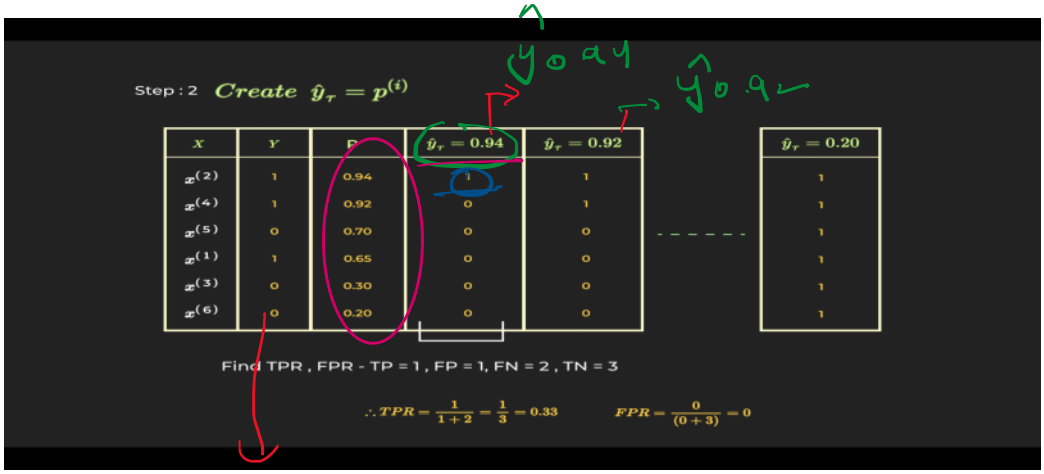


0.92 Threshold

≥ 0.92 → (1)

↳ 0

Threshold

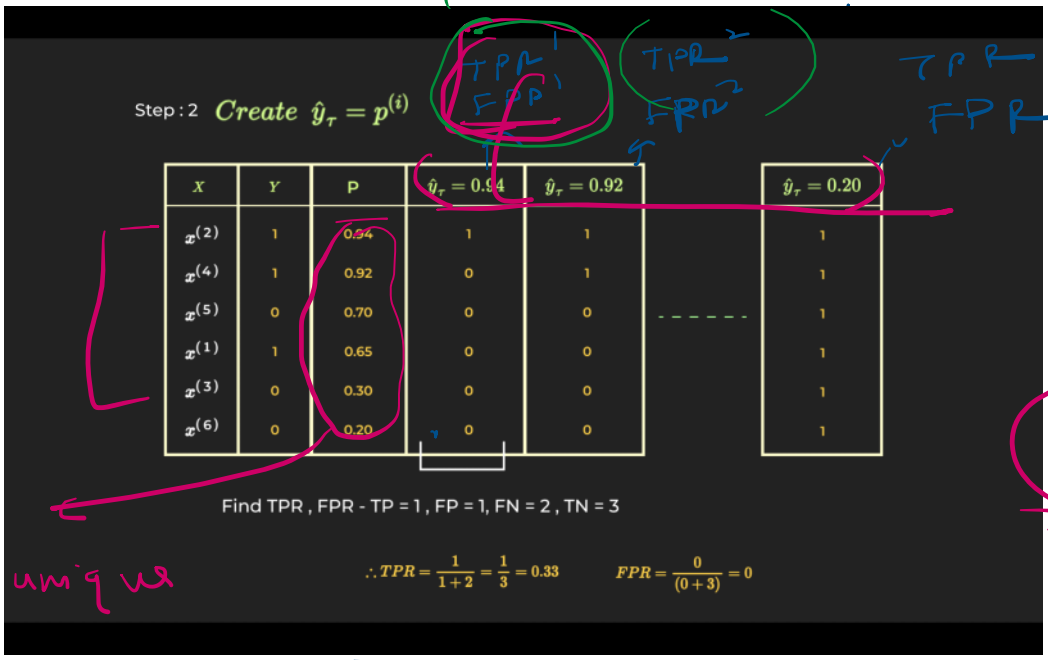


actual

TPR & FPR

TPR & FPR

FPR



TPR 6

FPR 6

scenario

plot

6 unique

TPR & FPR → 6 points

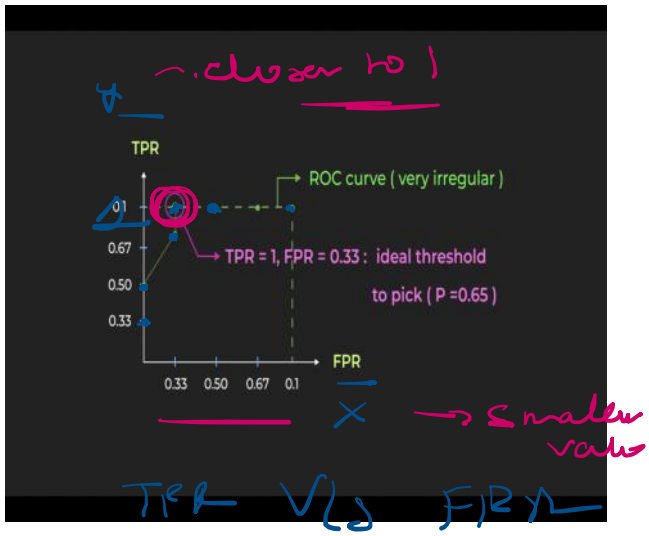
Repeating step - 2 for each P :

P	TPR	FPR
0.94	0.33	0
0.92	0.50	0
0.70	0.67	0.33
0.65	1	0.33
0.30	1	0.67
0.20	1	1

Pair of TPR & FPR for all 6 probabilities

Threshold

Plot



closer to 1

smaller value

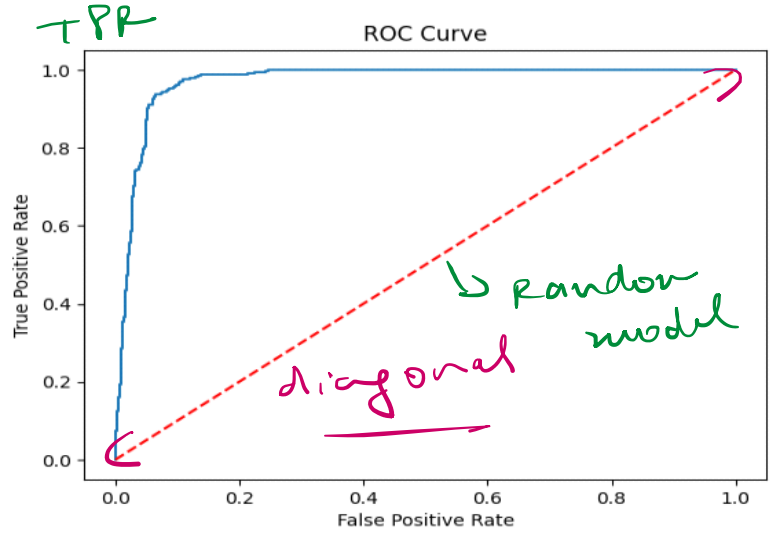
TPR vs FPR

⇒ FPR ↓ TPR ↑

goal

ROC process

⇒ TPR ↑ , FPR ↓



TPR

random model

diagonal

Red line → Random model

Heads → (1,0)

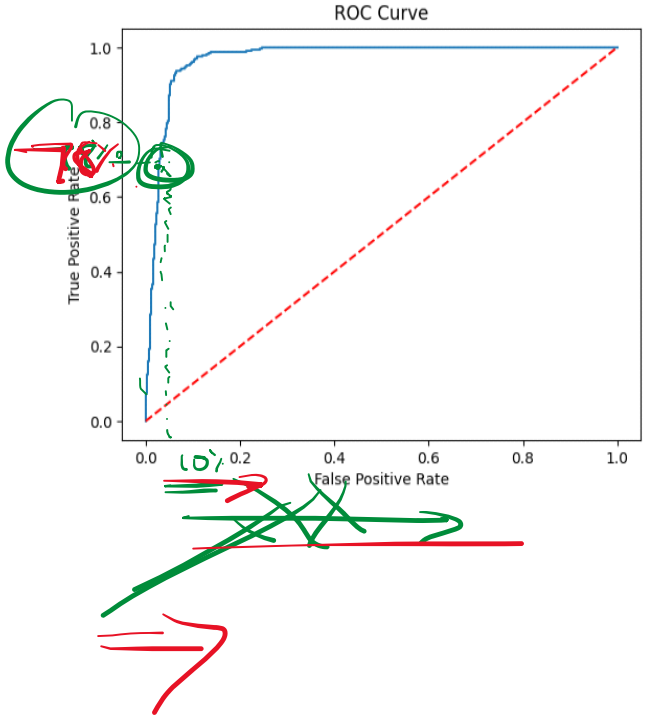
Tail → (0,1)

FPR

Next

False Positive Rate

ROC curve →



STRICT

→ ML algo

M_A M_B

Same problem

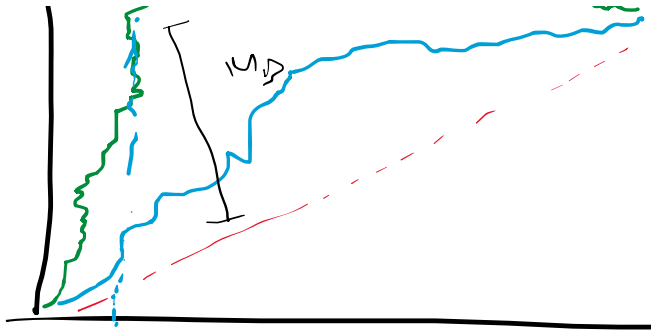
TP ↑

FPR ↓

Better model

which is Better ?

FPR



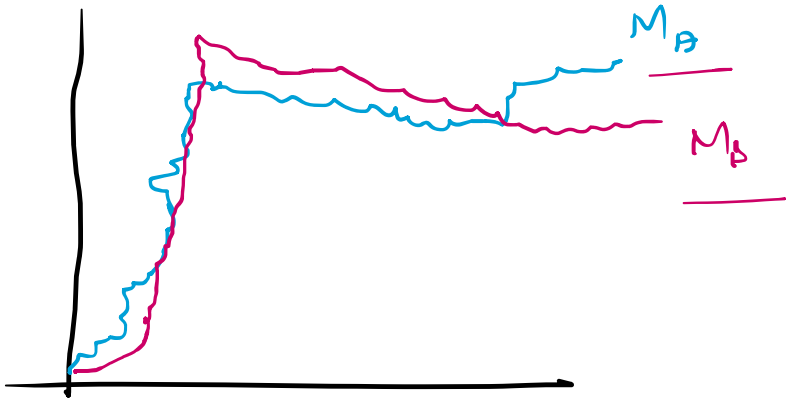
Better ?

good

ROC

Quantitatively

quantity_value

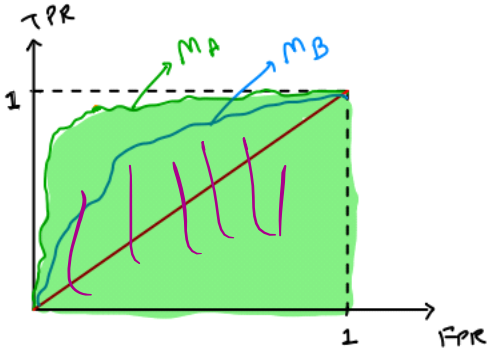


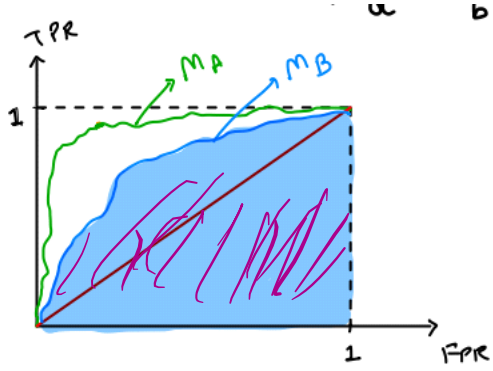
Sometimes MA MB

Term quantity

AUC → Area under curve

AU ROC →

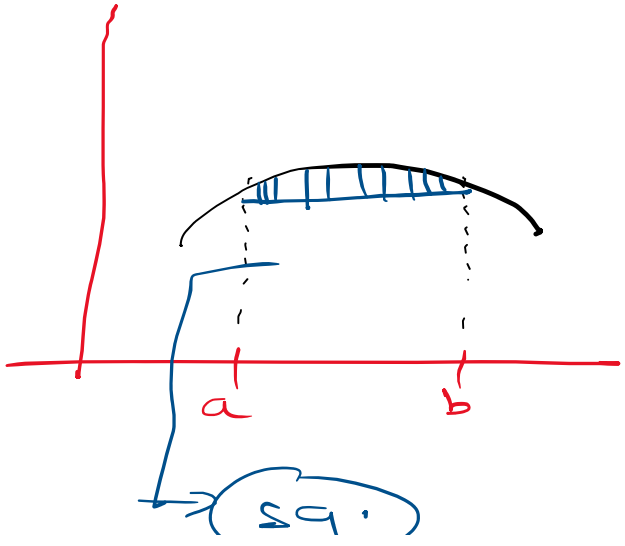




(X) AUC M_A ? AUC M_B

IntegRation

Calculs → MD X



X

→ smaller shapes ∫ computing area

(sq.

area

sq.

⇒

Trapezoid Rule

Sklearn → ROC_auc
– slope

④ SHORTCOMing with AUC ROC

① Binary classes →

② imbalance →

③ ordering of prob. & not actual prob







ROC process

ROC

TPR

FPR

TPR

FPR

$t = 0.95$

1
0
0
0
0

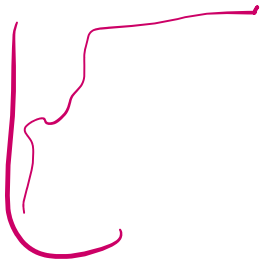
$t = 0.2$

1
0
0
0
0

TPR, FPR

TPR, FPR





Breaks

overcome →

10:30

unique pred.

Quiz time!

Time Left: 22s

How many points are typically used to plot an ROC curve?

1 user has participated

A	2 points (0,0) and (1,1)	0%
B	3 points representing the thresholds 0.25, 0.5, and 0.75	0%
C	10 points equally spaced between 0 and 1	100%
D	Depends on the number of unique threshold values	0%

End Quiz Now

Quiz time!

Time Left: 0s

Which of the following metrics can be directly derived from the ROC curve?

25 users have participated

A	Accuracy	4%
B	Precision	4%
C	Recall	4%
D	Area Under the Curve (AUC)	88%

End Quiz Now

overcome → PR CURVE

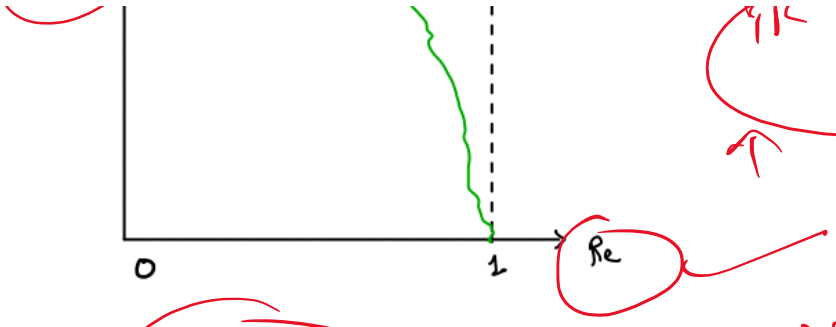
Precision Recall curve ⇒

ROC ⇔ PR

Approx

PR 1

PR ↑ CURVE



Plot

PR curve

TPR → Replaced By

FPR

Precision & recall

ROC → PR curve

Threshold

Imbalanced DATA ⇒

50 – 50 → Balanced

60 – 40 → Almost Balanced

70 – 30 → imBalance

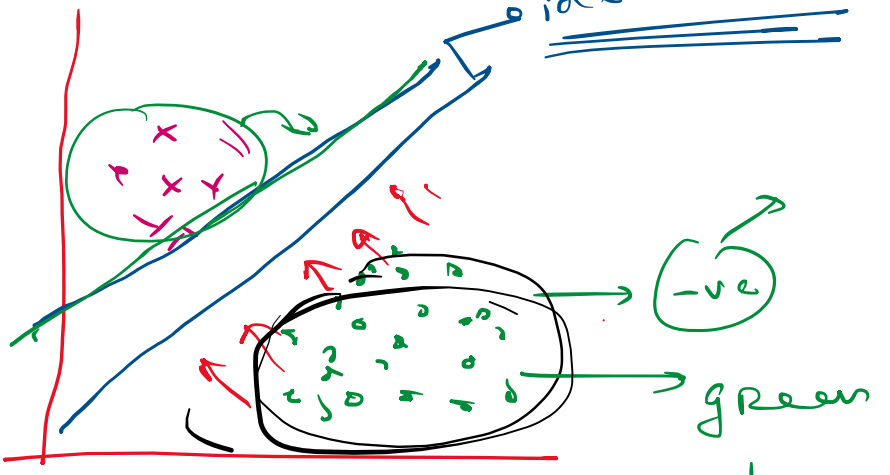
(*) 70 – 30 → imBalance

80 – 20 → Highly imB

95 – 5 → Extremely imB.

(+)

(+) impact of imbalanced DATA on your model



High dominance

LR model → 87%.

(1) →

assign wt. to minority class

Minority → more wt.

Minority → more

1000

Spam → 150

NS → 850

850 / 150 = 85 : 15

$$w = \frac{85}{15} = 5.6$$

150 × 5.6 = 850

1 Spam = 5.6 NS points

more power

Spam → wt = 5.6

NS → wt = 1

class_weight : {0 : 1, 1 : 5.67}

⇒ for every DP → finding loss

$$loss = \sum_{i=1}^{n} \log loss_i \times w_i$$

$w_i = 1 \rightarrow$ NS

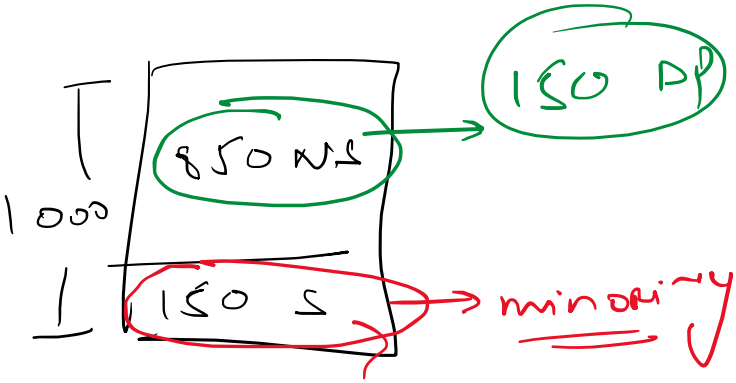
$w_i = 5.6 \rightarrow$ Spam

⇒ assigning wt.

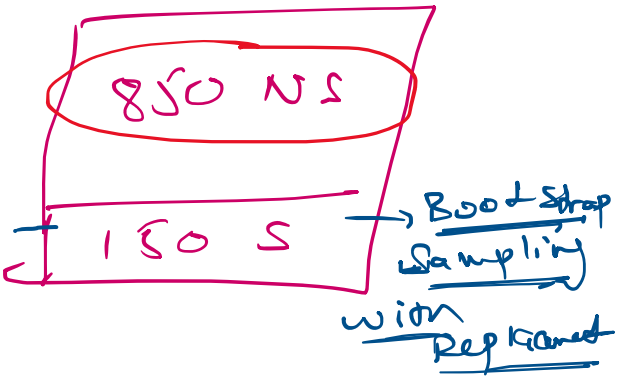
X

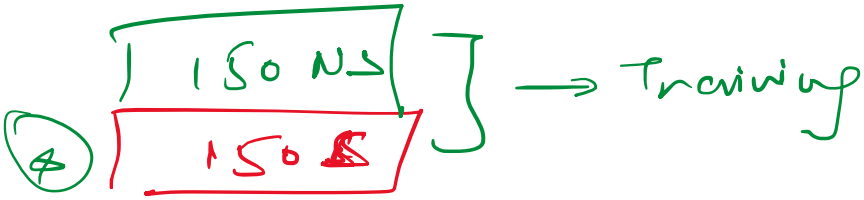
Undersampling | OVERsampling

Undersampling

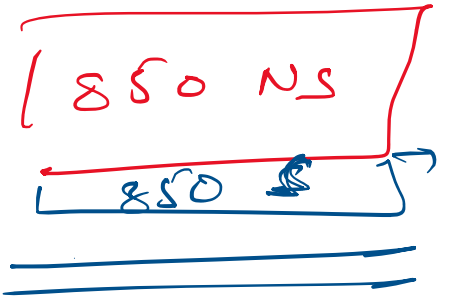


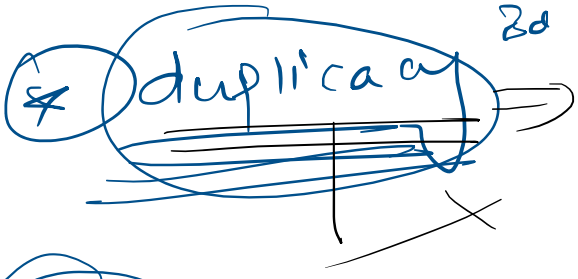
OVERSampling



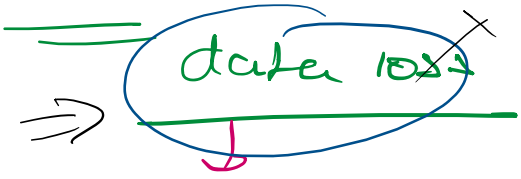


↳ Balanced





Problem →



Not Happy



Overfitting



Duplicacy

larger DATA

DATA LOSS

Data Augmentation

→ Deep learning

1 img → 10 img

Same idea → (MC)

SMOTE → Synthetically oversampling Estimation

minority Technique

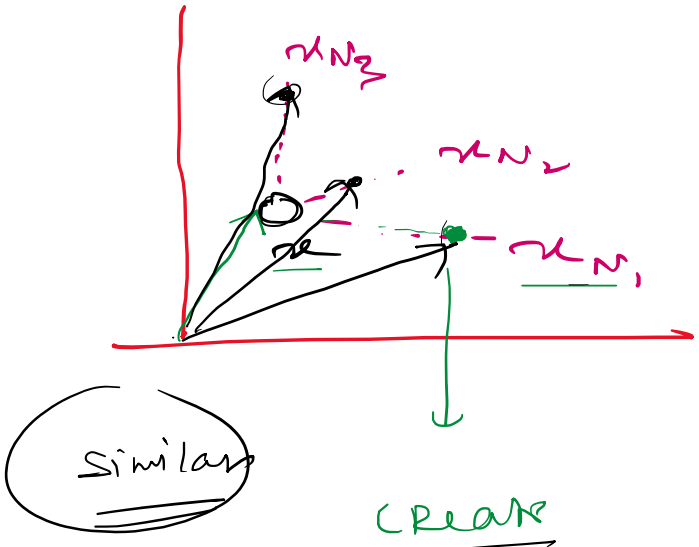
Synthetic Data →

Generating DP → similar

Minority class →

(1) select any Random point

(2) find k = 3 Nearest points



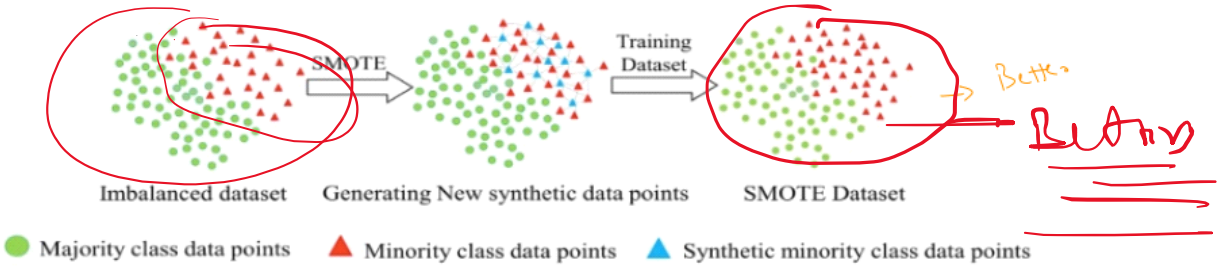
CREATE new DP vector

(3) α = Random (0, 1)

α → 0.3, 0.2, 0.6 ✓

$x_{new_1} = x + \alpha \cdot d(x_1, x_{N_1})$

$x_{new_2} = x + \alpha \, d(x_1, x_{N_2})$



SMOTE

& class wt.

Quiz time!

Time Left: 38s

Why are class weights important in imbalanced datasets?

1 user has participated

A	They improve the convergence speed of the model	0%
B	They reduce overfitting in the model	0%
✓ C	They help give more importance to the minority class	100%
D	They enhance the model's ability to handle missing values	0%

End Quiz Now

Fraud Detection

10k Txns → 100 → Fraud

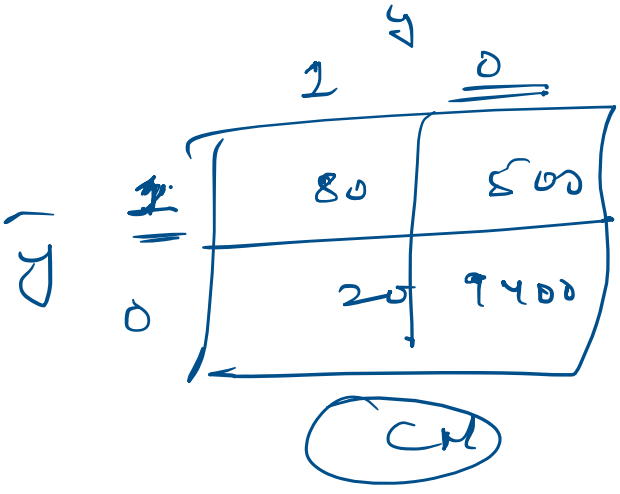
→ 9900 → Non

→ Fraud

→ imBalanced

MA

ROC



$TPR = \frac{80}{100} = 0.8$

$FPR = \frac{500}{9900}$

$= 0.050$

5% FPR

fantastic

80 → True frauds

500 → false alarms

Precision = $\frac{80}{20 + 500} = \frac{80}{580}$

